

[BUG] Possible bug in Hamiltonian observable unflattening when using `ja`

<https://github.com/PennyLaneAI/pennylane/issues/4601>

ROW 31

[BUG] Possible bug in Hamiltonian observable unflattening when using `jax.jit` #4601

New issue

Closed

#4601



erick-xanadu opened on Sep 15, 2023 · edited by erick-xanadu

Edits ...

Expected behavior

```
import pennylane as qml
from pennylane import numpy as pnp
import numpy as np
import pytest
import jax

class TestBroadcastExpand:
    def test_expansion_qnode(self):

        coeffs0 = [0.3, -5.1]
        obs = qml.Hamiltonian(qml.math.array(coeffs0), [qml.PauliZ(0), qml.PauliY(1)])
        @qml.qnode(qml.device("default.qubit", wires=2), interface="jax")
        def circuit(obs):
            return qml.state()

        expected = circuit(obs)
        print(expected)
```

Should be compatible with `jax.jit`

Actual behavior

Unflattening observable Hamiltonians when `jax.jit` ing a function raises an exception.

```
import pennylane as qml
from pennylane import numpy as pnp
import numpy as np
import pytest
import jax

class TestBroadcastExpand:
    def test_expansion_qnode(self):

        coeffs0 = [0.3, -5.1]
        obs = qml.Hamiltonian(qml.math.array(coeffs0), [qml.PauliZ(0), qml.PauliY(1)])
        @qml.qnode(qml.device("default.qubit", wires=2), interface="jax")
        def circuit(obs):
            return qml.state()

        expected = circuit(obs)
        print(expected)
        observed = jax.jit(circuit)(obs) # <-- extra line causes an exception
```

Additional information

I run this file with `pytest`.

Source code

```
import pennylane as qml
from pennylane import numpy as pnp
import numpy as np
import pytest
import jax

class TestBroadcastExpand:
    def test_expansion_qnode(self):
```

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

improve builtins shape and ndim; add `'math.get_deep_interface'`
PennyLaneAI/pennylane

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



```
def test_expansion_qnode(self):

    coeffs0 = [0.3, -5.1]
    obs = qml.Hamiltonian(qml.math.array(coeffs0), [qml.PauliZ(0), qml.PauliY(1)])
    @qml.qnode(qml.device("default.qubit", wires=2), interface="jax")
    def circuit(obs):
        return qml.state()

    expected = circuit(obs) # <-- succeeds
    print(expected)
    observed = jax.jit(circuit)(obs) # <-- raises exception
```

Tracebacks

```
env/lib/python3.10/site-packages/jax/_src/pjit.py:253: in cache_miss
    outs, out_flat, out_tree, args_flat, jaxpr = _python_pjit_helper(
env/lib/python3.10/site-packages/jax/_src/pjit.py:161: in _python_pjit_helper
    args_flat, _, params, in_tree, out_tree, _ = infer_params_fn(
env/lib/python3.10/site-packages/jax/_src/api.py:324: in infer_params
    return pjit.common_infer_params(pjit_info_args, *args, **kwargs)
env/lib/python3.10/site-packages/jax/_src/pjit.py:491: in common_infer_params
    jaxpr, consts, canonicalized_out_shardings_flat = _pjit_jaxpr(
env/lib/python3.10/site-packages/jax/_src/pjit.py:969: in _pjit_jaxpr
    jaxpr, final_consts, out_type = _create_pjit_jaxpr(
env/lib/python3.10/site-packages/jax/_src/linear_util.py:345: in memoized_fun
    ans = call(fun, *args)
env/lib/python3.10/site-packages/jax/_src/pjit.py:922: in _create_pjit_jaxpr
    jaxpr, global_out_avals, consts = pe.trace_to_jaxpr_dynamic(
env/lib/python3.10/site-packages/jax/_src/profiler.py:314: in wrapper
    return func(*args, **kwargs)
env/lib/python3.10/site-packages/jax/_src/interpreters/partial_eval.py:2155: in trace_to_jaxpr_dynamic
    jaxpr, out_avals, consts = trace_to_subjaxpr_dynamic(
env/lib/python3.10/site-packages/jax/_src/interpreters/partial_eval.py:2177: in trace_to_subjaxpr_dynamic
    ans = fun.call_wrapped(*in_tracers_)
env/lib/python3.10/site-packages/jax/_src/linear_util.py:183: in call_wrapped
    args, kwargs = next(gen)
env/lib/python3.10/site-packages/jax/_src/api_util.py:86: in flatten_fun_nokwargs
    py_args = tree_unflatten(in_tree, args_flat)
env/lib/python3.10/site-packages/jax/_src/tree_util.py:93: in tree_unflatten
    return treedef.unflatten(leaves)
../pennylane/pennylane/pytrees.py:35: in jax_unflatten
    return unflatten_fn(parameters, aux)
../pennylane/pennylane/ops/qubit/hamiltonian.py:178: in _unflatten
    new_op = cls(data[0], data[1])
../pennylane/pennylane/ops/qubit/hamiltonian.py:191: in __init__
    if qml.math.shape(coeffs)[0] != len(observables):
env/lib/python3.10/site-packages/autoray/autoray.py:79: in do
    return get_lib_fn(backend, fn)(*args, **kwargs)
../pennylane/pennylane/math/single_dispatch.py:39: in <lambda>
    ar.register_function("builtins", "shape", lambda x: np.array(x).shape)
-----

self = Traced<ShapedArray(float32[1])>with<DynamicJaxprTrace(level=1/0)>, args = (), kw = {}

    def __array__(self, *args, **kw):
>       raise TracerArrayConversionError(self)
E       jax._src.traceback_util.UnfilteredStackTrace: jax.errors.TracerArrayConversionError: The numpy.ndarray c
E       The error occurred while tracing the function circuit at /home/erick.choalopez/Code/catalyst-latest/bug
E       See https://jax.readthedocs.io/en/latest/errors.html#jax.errors.TracerArrayConversionError
E
E       The stack trace below excludes JAX-internal frames.
E       The preceding is the original exception that occurred, unmodified.
E
E       -----

env/lib/python3.10/site-packages/jax/_src/core.py:611: UnfilteredStackTrace

The above exception was the direct cause of the following exception:

self = <bug.TestBroadcastExpand object at 0x7f6f3e85b4f0>

    def test_expansion_qnode(self):

        coeffs0 = [0.3, -5.1]
        obs = qml.Hamiltonian(qml.math.array(coeffs0), [qml.PauliZ(0), qml.PauliY(1)])
        @qml.qnode(qml.device("default.qubit", wires=2), interface="jax")
        def circuit(obs):
            return qml.state()

        expected = circuit(obs)
        print(expected)
>       observed = jax.jit(circuit)(obs)
```

```

        expected = circuit(obs)
        print(expected)
>         observed = jax.jit(circuit)(obs)

bug.py:18:
-----
../pennylane/pennylane/pytrees.py:35: in jax_unflatten
    return unflatten_fn(parameters, aux)
../pennylane/pennylane/ops/qubit/hamiltonian.py:178: in _unflatten
    new_op = cls(data[0], data[1])
../pennylane/pennylane/ops/qubit/hamiltonian.py:191: in __init__
    if qml.math.shape(coeffs)[0] != len(observables):
env/lib/python3.10/site-packages/autoray/autoray.py:79: in do
    return get_lib_fn(backend, fn)(*args, **kwargs)
-----

x = (Traced<ShapedArray(float32[])>with<DynamicJaxprTrace(level=1/0)>, Traced<ShapedArray(float32[])>with<DynamicJaxprTrace(level=1/0)>)

>   ar.register_function("builtins", "shape", lambda x: np.array(x).shape)
E   jax.errors.TracerArrayConversionError: The numpy.ndarray conversion method __array__() was called on traced array of type TracedShapedArray(float32[])
E   The error occurred while tracing the function circuit at /home/erick.ochoalopez/Code/catalyst-latest/bug.py:18:10
E   See https://jax.readthedocs.io/en/latest/errors.html#jax.errors.TracerArrayConversionError

../pennylane/pennylane/math/single_dispatch.py:39: TracerArrayConversionError

```

System information

Name: PennyLane
Version: 0.33.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/PennyLaneAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /home/erick.ochoalopez/Code/catalyst-latest/env/lib/python3.10/site-packages
Editable project location: /home/erick.ochoalopez/Code/pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, sympy
Required-by: pennylane-catalyst, PennyLane-Lightning

Platform info: Linux-5.15.0-83-generic-x86_64-with-glibc2.31
Python version: 3.10.11
Numpy version: 1.23.5
Scipy version: 1.10.0

Installed devices:

- default.gaussian (PennyLane-0.33.0.dev0)
- default.mixed (PennyLane-0.33.0.dev0)
- default.qubit (PennyLane-0.33.0.dev0)
- default.qubit.autograd (PennyLane-0.33.0.dev0)
- default.qubit.jax (PennyLane-0.33.0.dev0)
- default.qubit.legacy (PennyLane-0.33.0.dev0)
- default.qubit.tf (PennyLane-0.33.0.dev0)
- default.qubit.torch (PennyLane-0.33.0.dev0)
- default.qutrit (PennyLane-0.33.0.dev0)
- null.qubit (PennyLane-0.33.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.32.0)

Using this commit-id for my version of pennylane, but this issue is likely unrelated to the commit below.

cbc8d19e6a36f0d4d38b44eabd4a8b98db677327

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



erick-xanadu added on Sep 15, 2023

Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/PennyLaneAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /home/erick.ochaalopez/Code/catalyst-latest/env/lib/python3.10/site-packages
Editable project location: /home/erick.ochaalopez/Code/pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: pennylane-catalyst, PennyLane-Lightning

Platform info: Linux-5.15.0-83-generic-x86_64-with-glibc2.31
Python version: 3.10.11
Numpy version: 1.23.5
Scipy version: 1.10.0
Installed devices:
- default.gaussian (PennyLane-0.33.0.dev0)
- default.mixed (PennyLane-0.33.0.dev0)
- default.qubit (PennyLane-0.33.0.dev0)
- default.qubit.autograd (PennyLane-0.33.0.dev0)
- default.qubit.jax (PennyLane-0.33.0.dev0)
- default.qubit.legacy (PennyLane-0.33.0.dev0)
- default.qubit.tf (PennyLane-0.33.0.dev0)
- default.qubit.torch (PennyLane-0.33.0.dev0)
- default.qutrit (PennyLane-0.33.0.dev0)
- null.qubit (PennyLane-0.33.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.32.0)

Using this commit-id for my version of pennylane, but this issue is likely unrelated to the commit below.

cbc8d19e6a36f0d4d38b44eabd4a8b98db677327

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



erick-xanadu added **bug** on Sep 15, 2023



timmysilv on Sep 15, 2023

Contributor ...

thanks for reporting this! it's caused by `[jnp.array(0.1), jnp.array(0.2)]` being treated as a builtins type (`list`). It's kind of a known thing that we avoid on our own, and it's especially troublesome with Hamiltonian coeffs. I've started a fix PR in [#4603](#), and I'll add some more testing before merging. It's not a totally generalized fix, but it'll cover most cases (so far, it wouldn't cover things like `[[jnp.array(1)]]`)



timmysilv mentioned this on Sep 18, 2023

[improve builtins shape and ndim; add `math.get\_deep\_interface` #4603](#)

timmysilv closed this as **completed** in [#4603](#) on Oct 3, 2023

timmysilv added a commit that references this issue on Oct 3, 2023

`improve builtins shape and ndim; add math.get_deep_interface` ([#4603](#))

Verified d799923

improve builtins shape and ndim; add `math.get_deep_interface` #4603

[Code](#)

Merged timmysilv merged 9 commits into `master` from `better-builtins-ndim-shape` on Oct 3, 2023

Conversation 5 Commits 9 Checks 0 Files changed 5

+74 -3



timmysilv commented on Sep 15, 2023 · edited

Contributor

Context:

Sometimes people do things on lists like `[jnp.array(1.1), jnp.array(2.2)]` and autoray thinks the interface is `builtins`. This happens a lot with `Hamiltonian.coeffs`. I'm here to help autoray.

Description of the Change:

- Add `qml.math.get_deep_interface` to extract out the interface from one of those (arbitrarily) deeply nested lists of lists, or tuples of tuples. Note that I assume the struct isn't ragged, because if you're asking for the interface, you're probably about to cast the whole list and that can fail if it needs to
- Change the `builtins` implementation for `ndim` and `shape` to check if it's a nested type containing a different interface. Then, cast to that interface and use *its* implementation for `ndim` / `shape`

Benefits:

We can get the `ndim` / `shape` of ugly nested things, and the helper function might be generally helpful elsewhere

Possible Drawbacks:

None!!!

Related GitHub Issues:

Fixes [#4601](#)



Reviewers

[mudit2812](#)



[albi3ro](#)



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[\[BUG\] Possible bug in Hamiltonian observabl...](#)

Notifications

Customize

[Subscribe](#)

You're not receiving notifications from this thread.

[improve builtins shape and ndim](#)

1010fe0

[timmysilv](#) mentioned this pull request on Sep 15, 2023

[\[BUG\] Possible bug in Hamiltonian observable unflattening when using `jax.jit` #4601](#)

[Closed](#)

1 task